

Solution Requirements

Date	15 March 2026
Team ID	SWTID-2026-8154
Project Name	Fitbuddy
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	Login with email and password; validate inputs and protect passwords. Provide sign-out.	High
2	Authorization levels	Single user role: each user can view and manage only their own profile and fitness plans.	High
3	External interfaces	Connect to Google Gemini API to generate personalized fitness plans; require a valid API key.	High

		Validate user inputs and safely send profile details to the AI service; display generated plan.	High
4	Transactions processing	Show workout schedules, exercise details, duration and progress summary; allow viewing saved plans.	Medium
5	Reporting	Generate plans based on user goals, fitness level, available time and equipment; avoid unsupported medical advice.	High
6	Business rules, etc.	Use the Gemini API and handle personal data securely; provide a notice that plans are informational, not medical advice.	High
7	Compliance to laws or regulations	Allow users to edit profile details, regenerate a plan and view their fitness recommendations.	Medium
8	Other	<i>[Add any additional functional requirements here]</i>	Low

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria

1	Performance & Speed	Plan generation and main screens should respond promptly under normal network conditions.	Target: screen loads ? 3 seconds; plan response ? 30 seconds, subject to API.
2	Scalability	Support growth in registered users and saved plans without redesigning the application.	Acceptance: data remains available as usage increases; API limits handled gracefully.
3	Security & Data Privacy	Protect API key on the server; validate inputs and avoid exposing private profile data.	Acceptance: secrets stored in environment variables; HTTPS used in deployment.
4	Reliability & Availability	Show a clear error and allow retry if the AI service or network is unavailable.	Acceptance: no app crash; retry option displayed.
5	Usability & Accessibility	Use simple forms, readable labels and responsive layout for desktop and mobile.	Acceptance: core flow usable on common mobile and desktop screen sizes.
6	<i>Other</i>	Keep frontend, backend and AI integration modular for easier updates and deployment.	Acceptance: setup documented; configuration via environment variables.